

CAMPUSATTACH

Student Attachment and Internship Opportunity Platform

TASK 1: PROJECT PLANNING AND REQUIREMENTS ANALYSIS

Student: Kyungu Dennis Muuo

Programme: CDAM Software Development Internship

Project: CampusAttach

Task: Task 1 — Project Planning and Requirements Analysis

1. Identify a Real-World Problem

University students often experience difficulties when searching for attachment and internship opportunities during their studies.

Information about available opportunities may be distributed across different channels, including social media platforms, WhatsApp groups, university notices, personal networks, company websites, and other online sources. As a result, students may need to search through several different sources before finding opportunities that are relevant to their course, skills, preferred location, and attachment period.

Students may also experience difficulties understanding application requirements, identifying deadlines, and keeping track of applications they have already submitted.

This problem is personally relevant to the project because, during my three years in university, I have observed my brother and friends struggling to find suitable attachment and internship opportunities. I will also need to look for an attachment opportunity during my studies within approximately the next eight months.

Based on this experience, I identified an opportunity to develop CampusAttach, a web-based platform intended to organize the process of discovering and managing attachment and internship opportunities.

Problem Context

The first version of CampusAttach will focus on providing students with a structured way to discover opportunities and manage their applications, while providing organizations with a way to publish opportunities and manage applications.

The system is intended to improve the organization and accessibility of opportunity information. It will not guarantee that a student receives an attachment or internship placement.

2. Problem Statement

University students need attachment and internship opportunities to gain practical experience and meet academic or career-related requirements. However, information about available opportunities is often scattered across multiple sources.

Students may therefore spend considerable time searching through different platforms, social media groups, university notices, personal contacts, and organization websites. This can make it difficult to identify suitable opportunities, understand application requirements, identify deadlines, and keep track of application progress.

Organizations may also lack a structured platform specifically focused on connecting available attachment and internship opportunities with students who are looking for placements.

The absence of a centralized platform for discovering and managing attachment and internship opportunities can contribute to repeated searches, missed deadlines, incomplete information, and difficulty tracking applications.

There is therefore a need for a web-based platform that brings relevant opportunity information into one organized system and provides students with tools for discovering opportunities, viewing details, applying, and tracking application status. The platform should also provide organizations with a structured way to publish opportunities and manage applications.

3. Project Objectives

3.1 General Objective

To develop a web-based platform that helps university students discover and manage attachment and internship opportunities while providing organizations with a structured way to publish and manage such opportunities.

3.2 Specific Objectives

1. To provide students with accounts and profiles containing relevant academic and contact information.
2. To allow students to browse available attachment and internship opportunities.
3. To provide search and filtering functionality to help students find relevant opportunities.
4. To allow students to view complete opportunity details, requirements, and deadlines before applying.
5. To allow students to submit applications for opportunities that support applications through the platform.
6. To allow students to view and track the status of their submitted applications.
7. To provide organizations with accounts through which they can create and manage opportunity listings.
8. To allow organizations to review applications received for their opportunities and update application status.
9. To provide a system administrator with tools for managing users and moderating opportunity listings.

10. To maintain student, organization, opportunity, and application information in an organized and secure system.

4. Stakeholders

The following stakeholders have been identified for CampusAttach:

4.1 University Students

Students are the primary users of CampusAttach. They will use the platform to create profiles, search for opportunities, view opportunity information, apply for available opportunities, and track their applications.

4.2 Organizations / Employers

Organizations will use the platform to publish attachment and internship opportunities, manage their listings, review applications, and update application statuses.

4.3 System Administrator

The system administrator will manage platform users, moderate opportunity listings, and perform other permitted administrative activities.

4.4 University Attachment or Career Support Personnel

University attachment or career support personnel may have an interest in the availability and organization of attachment opportunities and may support students in attachment-related activities.

4.5 University Attachment Supervisors or Coordinators

Attachment supervisors or coordinators may have an interest in student attachment processes and may support students in fulfilling attachment requirements.

4.6 Project Owner / Developer

The project owner and developer is responsible for analyzing the problem, designing the system, implementing the application, testing it, documenting it, and maintaining it.

5. Functional Requirements

Functional requirements describe the functions that CampusAttach should provide.

5.1 User Account and Authentication

FR-01: The system shall allow a student to create an account.

FR-02: The system shall allow an organization to create an account.

FR-03: The system shall allow registered users to log in.

FR-04: The system shall allow authenticated users to log out.

FR-05: The system shall restrict protected functions to authorized users.

5.2 Student Profile Management

FR-06: The system shall allow students to create their profiles.

FR-07: The system shall allow students to update their profiles.

FR-08: A student profile shall contain relevant information such as name, institution, course, year of study, skills, preferred location, and contact information.

5.3 Opportunity Management

FR-09: The system shall allow authorized organizations to create opportunity listings.

FR-10: An opportunity shall contain information such as title, organization, description, requirements, location, opportunity type, deadline, and application instructions.

FR-11: The system shall allow organizations to edit their own opportunity listings.

FR-12: The system shall allow organizations to manage their own posted opportunities.

FR-13: The system administrator shall be able to review and moderate opportunity listings.

5.4 Opportunity Discovery

FR-14: The system shall allow students to browse available opportunities.

FR-15: The system shall allow students to search for opportunities using keywords.

FR-16: The system shall allow students to filter opportunities using available criteria.

FR-17: The system shall allow students to view complete opportunity details.

5.5 Applications

FR-18: The system shall allow students to apply for opportunities where platform applications are enabled.

FR-19: The system shall record the student associated with an application.

FR-20: The system shall record the opportunity associated with an application.

FR-21: The system shall record the application date.

FR-22: The system shall record the application status.

FR-23: The system shall allow students to view their submitted applications.

FR-24: The system shall allow students to track their application status.

FR-25: The system shall allow organizations to view applications received for their own opportunities.

FR-26: The system shall allow organizations to update application status.

5.6 Administration

FR-27: The system administrator shall be able to view registered users.

FR-28: The system administrator shall be able to manage permitted user records.

FR-29: The system administrator shall be able to moderate opportunity listings.

FR-30: The system administrator shall be able to remove inappropriate or misleading opportunity content where permitted.

FR-31: The system administrator shall be able to monitor basic platform activity.

6. Non-Functional Requirements

Non-functional requirements describe the qualities and operating expectations of CampusAttach.

6.1 Usability

NFR-01: The interface should be simple enough for a first-time student user to understand.

NFR-02: Navigation should be consistent across the application.

NFR-03: Forms should provide clear validation messages.

NFR-04: The system should provide a responsive interface suitable for common laptop, tablet, and mobile screen sizes.

6.2 Performance

NFR-05: Common pages should load within a reasonable time under normal network conditions.

NFR-06: Search and filtering should return results efficiently for the expected initial number of records.

NFR-07: The backend should avoid unnecessary repeated database queries.

6.3 Security

NFR-08: User passwords must not be stored in plain text.

NFR-09: Authentication and authorization must be implemented for protected operations.

NFR-10: Users should only access information and actions permitted by their roles.

NFR-11: Sensitive configuration information such as database credentials must not be committed to GitHub.

NFR-12: Production communication should use HTTPS.

6.4 Privacy

NFR-13: Personal information should only be collected for legitimate CampusAttach purposes.

NFR-14: Access to personal information should be restricted according to user roles.

NFR-15: The system should handle stored personal information transparently and responsibly.

6.5 Reliability

NFR-16: The system should handle invalid input without crashing.

NFR-17: The system should provide useful error messages when an operation fails.

NFR-18: Database operations should maintain data consistency.

6.6 Maintainability

NFR-19: Source code should be organized into clear modules and components.

NFR-20: The system should use meaningful names and appropriate documentation.

NFR-21: Git should be used to maintain the project's version history.

6.7 Compatibility

NFR-22: The web application should work with modern browsers such as current versions of Chrome, Edge, and Firefox.

6.8 Scalability

NFR-23: The system design should allow additional students, organizations, opportunities, and application records to be added without requiring a complete redesign of the platform.

7. User Stories

7.1 Student User Stories

US-01: As a student, I want to create an account so that I can use CampusAttach.

US-02: As a student, I want to log in so that I can access my account and protected features.

US-03: As a student, I want to create and manage my profile so that relevant academic and contact information is available when needed.

US-04: As a student, I want to browse attachment and internship opportunities so that I can discover available placements.

US-05: As a student, I want to search for opportunities using keywords so that I can find relevant opportunities more quickly.

US-06: As a student, I want to filter opportunities so that I can reduce irrelevant results.

US-07: As a student, I want to view opportunity details and requirements so that I can decide whether an opportunity is suitable for me.

US-08: As a student, I want to apply for an opportunity so that I can submit my application.

US-09: As a student, I want to track my applications so that I can know the current status of each application.

7.2 Organization User Stories

US-10: As an organization, I want to create an account so that I can use CampusAttach to publish opportunities.

US-11: As an organization, I want to log in so that I can access my organization features.

US-12: As an organization, I want to create an opportunity listing so that students can discover available placements.

US-13: As an organization, I want to edit and manage my opportunity listings so that the information remains accurate.

US-14: As an organization, I want to view applications received for my opportunities so that I can review applicants.

US-15: As an organization, I want to update application status so that students can see the progress of their applications.

7.3 System Administrator User Stories

US-16: As a system administrator, I want to manage users so that the platform remains controlled.

US-17: As a system administrator, I want to moderate opportunity listings so that inappropriate or misleading content can be handled.

US-18: As a system administrator, I want to monitor basic platform activity so that I can support proper platform administration.

8. Use Case Descriptions

UC-01: Register Account

Primary Actor:

Student or Organization

Purpose:

To allow a new student or organization to create an account.

Preconditions:

The user does not already have an account using the same registered email address.

Main Flow:

1. The user selects Register.
2. The user selects the appropriate account type.
3. The user enters the required information.

4. The system validates the information.
5. The system creates the account.
6. The system confirms successful registration.

Alternative Flow:

If the information is invalid or the email is already registered, the system displays an appropriate error message and requests correction.

UC-02: Log In

Primary Actor:

Registered Student or Organization

Purpose:

To allow a registered user to access protected CampusAttach features.

Preconditions:

The user has an active account.

Main Flow:

1. The user enters their email and password.
2. The system validates the credentials.
3. The system authenticates the user.
4. The system grants access according to the user's role.
5. The system displays the appropriate user area.

Alternative Flow:

If the credentials are incorrect, the system displays an error message and does not grant access.

UC-03: Manage Student Profile

Primary Actor:

Student

Purpose:

To allow students to create and maintain their profile information.

Preconditions:

The student is authenticated.

Main Flow:

1. The student opens the profile section.
2. The student enters or edits profile information.
3. The system validates the information.
4. The system saves the changes.
5. The system confirms the update.

UC-04: Browse, Search and Filter Opportunities

Primary Actor:

Student

Purpose:

To help students find relevant attachment and internship opportunities.

Preconditions:

The system contains available opportunity listings.

Main Flow:

1. The student opens the opportunities section.
2. The student browses available opportunities.
3. The student enters a search keyword or selects available filters.
4. The system processes the search criteria.
5. The system displays matching opportunities.
6. The student selects an opportunity to view its details.

Alternative Flow:

If no opportunity matches the selected criteria, the system displays an appropriate message.

UC-05: View Opportunity Details

Primary Actor:

Student

Purpose:

To allow a student to view complete information about an opportunity before applying.

Preconditions:

The opportunity exists and is available for viewing.

Main Flow:

1. The student selects an opportunity.
2. The system retrieves the opportunity information.
3. The system displays the title, organization, description, requirements, location, opportunity type, deadline, and application instructions where available.
4. The student reviews the information.

UC-06: Apply for Opportunity

Primary Actor:

Student

Purpose:

To allow a student to submit an application for an available opportunity.

Preconditions:

The student is authenticated.

The opportunity is open for applications.

Main Flow:

1. The student opens an opportunity.
2. The student selects Apply.
3. The student provides or confirms the required application information.
4. The system validates the application.

5. The system records the application.
6. The system assigns the initial application status.
7. The system confirms successful submission.

Alternative Flow:

The system rejects an invalid or duplicate application or an application submitted after the opportunity has closed.

UC-07: Track Application

Primary Actor:

Student

Purpose:

To allow a student to view the current status of submitted applications.

Preconditions:

The student has submitted at least one application.

Main Flow:

1. The student opens My Applications.
2. The system retrieves the student's applications.
3. The system displays the applications and their current statuses.
4. The student selects an application to view additional information.

UC-08: Post Opportunity

Primary Actor:

Organization

Purpose:

To allow an organization to publish an attachment or internship opportunity.

Preconditions:

The organization is authenticated and authorized.

Main Flow:

1. The organization selects Post Opportunity.
2. The organization enters opportunity information.
3. The system validates the information.
4. The system stores the opportunity.
5. The opportunity becomes available according to the platform's moderation rules.

Alternative Flow:

If required information is missing or invalid, the system displays validation errors.

UC-09: Manage Posted Opportunities

Primary Actor:

Organization

Purpose:

To allow an organization to manage its own opportunity listings.

Preconditions:

The organization is authenticated and owns the relevant opportunity listing.

Main Flow:

1. The organization opens its opportunity management area.
2. The system displays the organization's listings.
3. The organization selects a listing.
4. The organization edits or manages the listing.
5. The system validates and saves the changes.

UC-10: Review Applicants and Update Status

Primary Actor:

Organization

Purpose:

To allow an organization to review applications and update their statuses.

Preconditions:

The organization has a posted opportunity with applications.

Main Flow:

1. The organization opens an opportunity.
2. The organization views received applications.

3. The organization selects an applicant.
4. The organization reviews the application information.
5. The organization updates the application status.
6. The system records the new status.

UC-11: Verify or Moderate Users and Opportunities

Primary Actor:

System Administrator

Purpose:

To help maintain appropriate users and opportunity content on the platform.

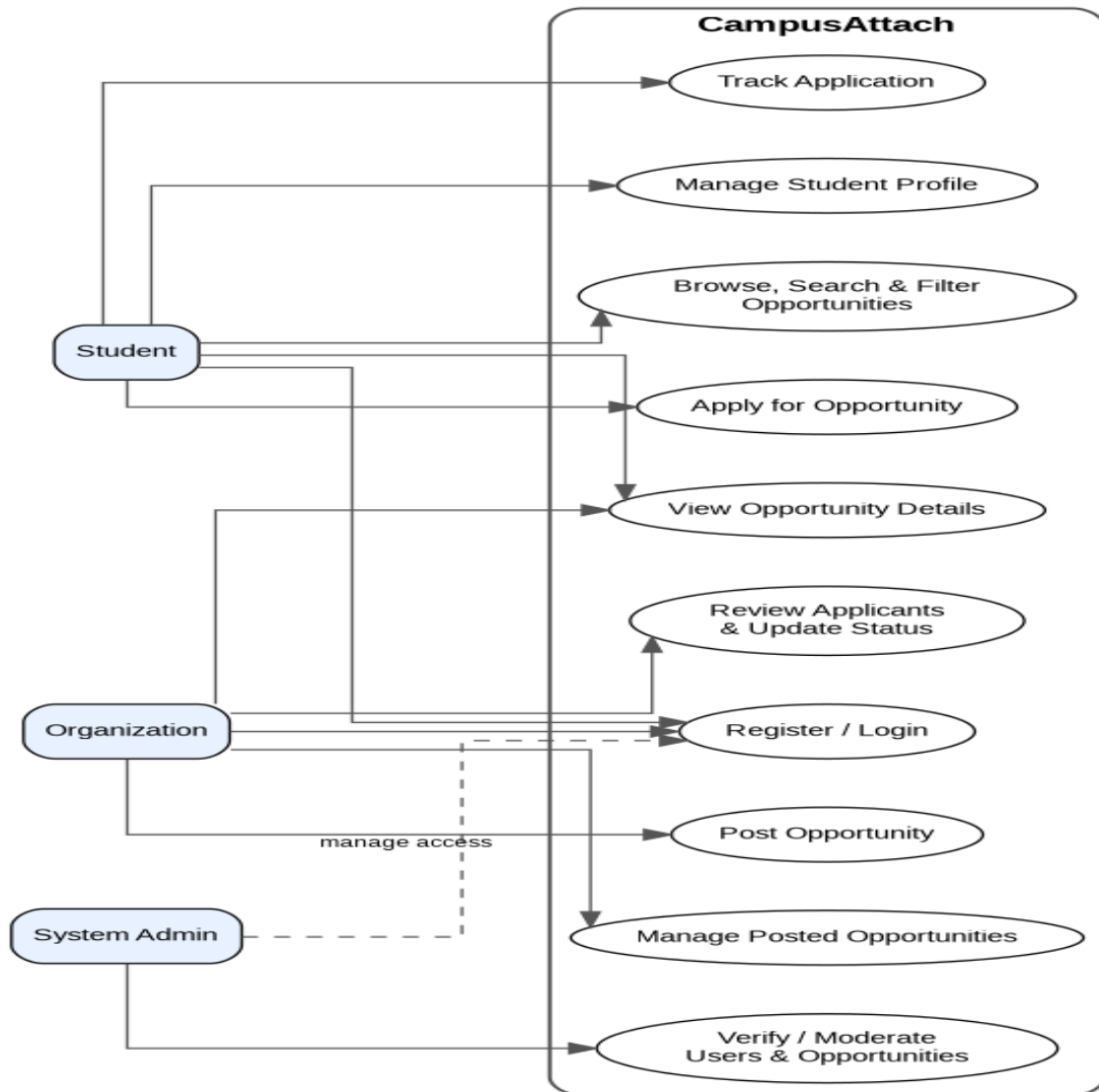
Preconditions:

The administrator is authenticated and authorized.

Main Flow:

1. The administrator opens the administration area.
2. The administrator reviews users or opportunity listings.
3. The administrator performs an allowed moderation action.
4. The system records the action.
5. The system updates the relevant record.

9. Use Case Diagram



10. Software Requirements Specification (SRS)

10.1 Purpose

This Software Requirements Specification defines the requirements for CampusAttach, a proposed web-based platform designed to help university students discover and manage attachment and internship opportunities.

The SRS provides a common description of the intended system functionality, users, data, security expectations, assumptions, constraints, and success criteria.

10.2 Scope

The first version of CampusAttach will focus on:

- User registration and authentication.
- Student profile management.
- Organization account management.
- Attachment and internship opportunity listings.
- Opportunity browsing.
- Opportunity search.
- Opportunity filtering.
- Opportunity detail viewing.
- Student applications.
- Application tracking.
- Organization application management.
- Basic administrator moderation.

The first version will not include advanced features such as AI-based matching, automated external website scraping, payment processing, or a dedicated mobile application unless such features are later approved as project enhancements.

10.3 Intended Users

The main intended users are:

Student:

Uses CampusAttach to create a profile, discover opportunities, view opportunity information, apply, and track applications.

Organization:

Uses CampusAttach to create and manage opportunity listings and review applications.

System Administrator:

Uses CampusAttach to manage users, moderate opportunity listings, and perform permitted administrative functions.

10.4 Product Perspective

CampusAttach is intended to operate as a web-based application.

The planned system will consist of:

- A browser-based frontend.
- A server-side backend API.
- A relational database.

The frontend and backend will communicate through RESTful APIs.

Detailed system architecture and database design will be developed during Task 2.

10.5 Planned Technology Environment

The selected technology environment for CampusAttach will follow the technologies provided by the CDAM internship guidelines.

Frontend:

- Angular
- TypeScript
- HTML5
- CSS3
- Tailwind CSS
- Angular Material where appropriate

Backend:

- NestJS
- TypeScript
- Prisma ORM

Database:

- PostgreSQL

Version Control:

- Git
- GitHub

API Testing:

- Postman

Planned Deployment:

- Frontend: Vercel or Netlify
- Backend: Render or Railway
- Database: Neon PostgreSQL or Supabase PostgreSQL

The exact versions and final deployment services will be documented when implementation begins.

10.6 System Interfaces

User Interface:

CampusAttach will provide a web interface through which students, organizations, and administrators interact with the system.

Application Interface:

The Angular frontend will communicate with the backend through RESTful API endpoints.

Database Interface:

The backend will communicate with the PostgreSQL database using Prisma ORM.

10.7 Data Requirements

The initial system is expected to store information relating to:

- User accounts.

- Student profiles.
- Organization profiles.
- Opportunity listings.
- Applications.
- Application statuses.
- Administrative records required for moderation.

Only information necessary for the intended functionality of CampusAttach should be collected.

10.8 Authentication and Authorization

CampusAttach shall use authentication for protected functions.

Role-based authorization shall separate the permissions of:

- Students.
- Organizations.
- System administrators.

Students should only manage their own profiles and applications.

Organizations should only manage their own opportunity listings and related applications.

Administrators should have access to permitted platform management and moderation functions.

10.9 Security and Privacy

The system shall follow responsible security practices.

These include:

- Passwords must not be stored in plain text.
- Passwords must be securely hashed.
- Protected operations must require authentication.
- Authorization checks must be applied to protected resources.
- Sensitive configuration values must be stored outside source code.
- Secrets must not be committed to GitHub.
- Production communication should use HTTPS.
- Personal information should only be collected for legitimate platform purposes.
- Access to personal information should be controlled according to user roles.

These requirements are particularly important because CampusAttach may handle student and organization personal information.

10.10 Assumptions

The following assumptions apply to the first version:

- Users will have access to an internet-connected device.
- Users will have a modern web browser.
- Students will provide accurate profile information.

- Organizations will provide accurate opportunity information.
- The initial system will operate as a web application.
- Organizations will be responsible for the accuracy of their opportunity information.
- The platform will organize opportunity discovery and application management but will not guarantee placement.

10.11 Constraints

The project has the following constraints:

- Development must be completed within the CDAM internship schedule.
- The system must follow the CDAM technical and documentation requirements.
- The project is being developed by a student intern.
- The first version must prioritize the core functionality required for a working demonstration.
- Infrastructure and hosting choices should remain practical for the project.
- The project must comply with academic-integrity and security requirements.

10.12 Business Rules

BR-01: A user must have an account before accessing protected features.

BR-02: Users may only perform actions allowed by their assigned roles.

BR-03: Students manage their own profiles and applications.

BR-04: Organizations manage their own opportunity listings.

BR-05: Organizations can manage applications associated with their own opportunities.

BR-06: Administrators can perform permitted moderation and platform-management activities.

BR-07: A student should not be allowed to submit a duplicate application for the same opportunity.

BR-08: Applications should not be accepted for opportunities that are no longer open.

10.13 Application Statuses

The initial application status flow may include:

- Pending
- Shortlisted
- Accepted
- Rejected

The organization will update the status and the student will be able to view the current status.

10.14 Error Handling

The system should provide clear feedback when:

- Required information is missing.

- User credentials are invalid.
- An unauthorized action is attempted.
- An opportunity cannot be found.
- An application is invalid.
- A duplicate application is submitted.
- A server or database operation fails.

The application should handle errors without exposing sensitive technical information to users.

10.15 Maintainability

The system should be organized into clear modules and components.

The source code should use meaningful names and consistent structure.

Git and GitHub will be used to maintain version history.

The project should contain appropriate documentation to make future maintenance easier.

10.16 Scalability

The design should allow the platform to grow to support additional:

- Students.
- Organizations.
- Opportunities.

- Applications.

The system should be designed so that future improvements can be added without completely redesigning the application.

10.17 Acceptance Direction

The core system should be considered functionally acceptable when:

- A student can register and log in.
- A student can manage a profile.
- A student can discover opportunities.
- A student can search and filter opportunities.
- A student can view opportunity details.
- A student can apply for an available opportunity.
- A student can track application status.
- An organization can register and log in.
- An organization can post and manage opportunities.
- An organization can review applications and update status.
- An administrator can perform basic moderation.
- Unauthorized actions are restricted.
- Data is stored and retrieved correctly.

Final testing and deployment will be completed during the later internship tasks.

11. How the Proposed System Addresses the Identified Problem

CampusAttach addresses the identified problem by bringing attachment and internship opportunity discovery and application management into one organized platform.

The relationship between the identified problems and the proposed system response is shown below.

Identified Problem:

Opportunity information is scattered across different sources.

Proposed Response:

CampusAttach provides a centralized platform where organizations can publish opportunities and students can discover them.

Identified Problem:

Students spend time searching through different sources for suitable opportunities.

Proposed Response:

CampusAttach provides browsing, keyword search, and filtering functionality.

Identified Problem:

Students may have difficulty understanding opportunity requirements and deadlines.

Proposed Response:

CampusAttach provides structured opportunity details containing information such as requirements, location, opportunity type, deadline, and application instructions.

Identified Problem:

Students may lose track of applications they have submitted.

Proposed Response:

CampusAttach provides a My Applications area where students can view submitted applications and their current statuses.

Identified Problem:

Organizations may not have a structured way to publish opportunities specifically for students seeking attachment or internship placements.

Proposed Response:

CampusAttach provides organization accounts and opportunity posting functionality.

Identified Problem:

Students may not know the progress of their applications.

Proposed Response:

Organizations can update application statuses and students can view the current status of their applications.

Identified Problem:

Inappropriate or misleading opportunity information may reduce trust in the platform.

Proposed Response:

CampusAttach provides a system administrator role for user and opportunity moderation.

Expected Impact

The proposed platform is intended to make the process of finding and managing attachment and internship opportunities more organized.

For students, the platform aims to reduce the effort involved in searching for opportunities, make important opportunity information easier to access, and improve application tracking.

For organizations, the platform aims to provide a structured way to publish opportunities and manage applications.

CampusAttach will not guarantee that a student receives an attachment or internship placement. Its purpose is to improve the organization and accessibility of opportunity discovery and application management.